Laboratory Work 6

Detecting Anomalies in Time-Series Data Using Power BI

IS107 | Power BI Desktop
Tourism Dataset Analysis

Overview

Dataset

LW6-tourism-details.csv — 920 rows of monthly tourism data (2012–2022)

Fields

Period (Date), Accommodation Type (Text), Travel (Text), Tourists (Number)

Tool

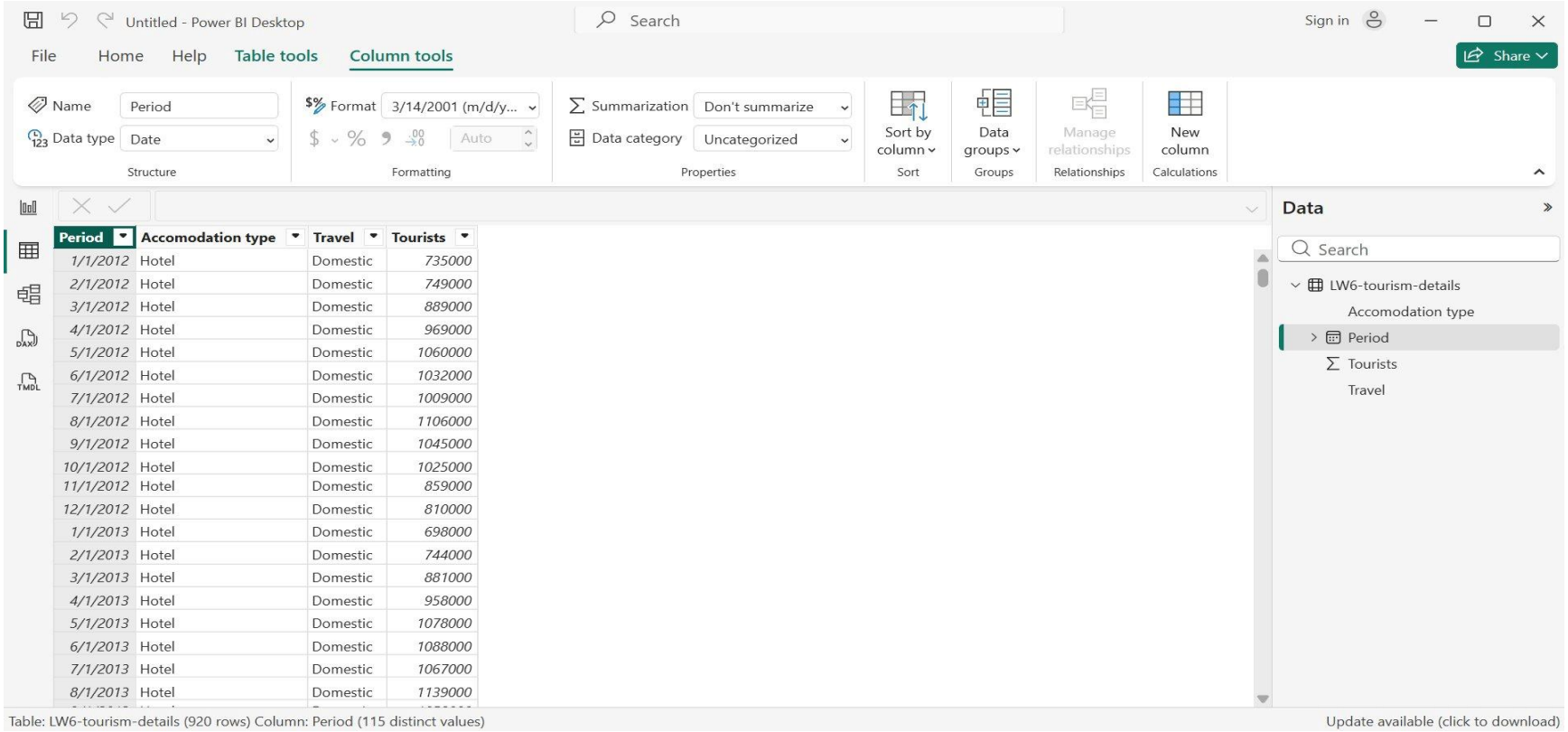
Power BI Desktop — Line Chart + Analytics Pane Anomaly Detection

Objective

Detect and explain anomalies in tourist volume over time

Part 1 — Data Preparation

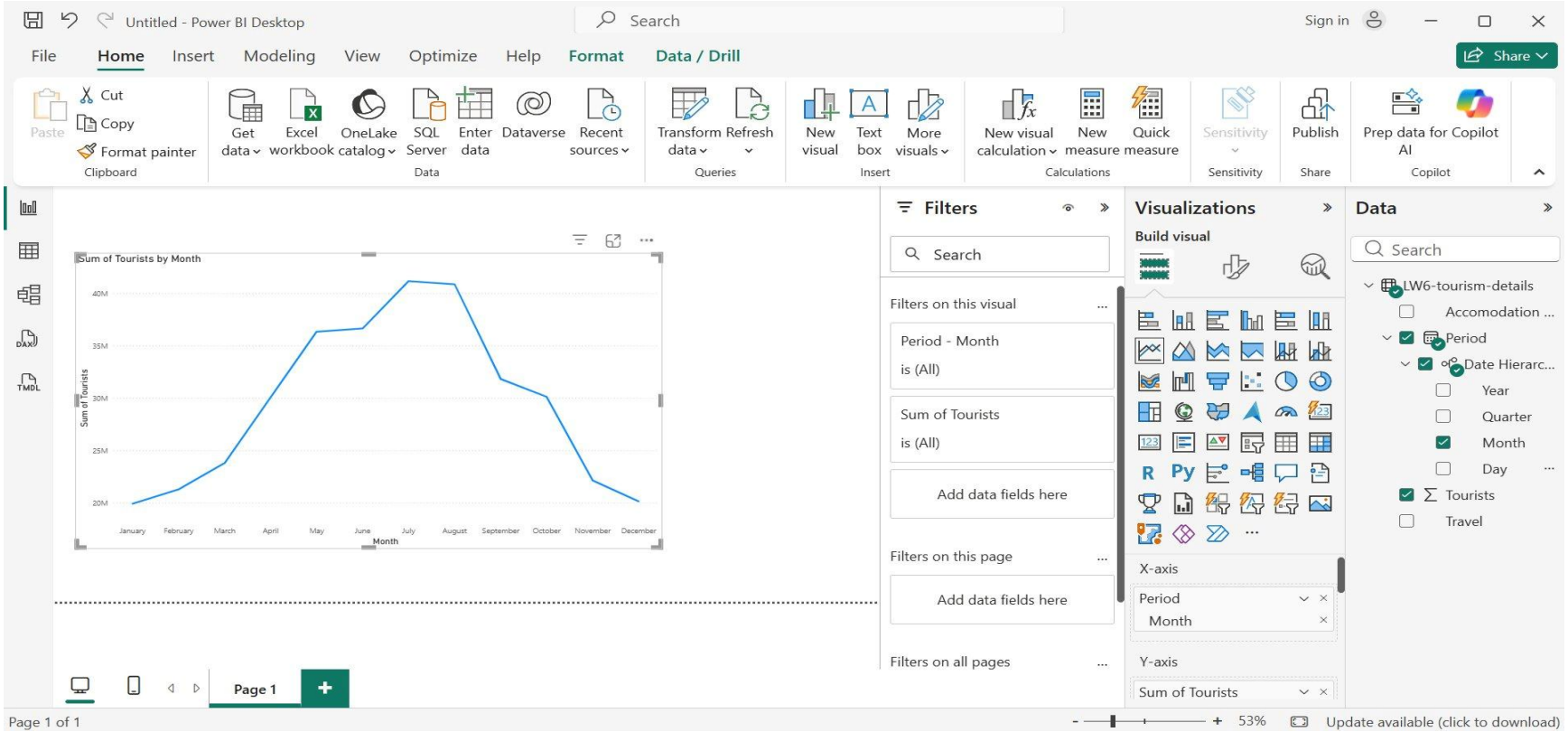
Imported LW6-tourism-details.csv into Power BI. The Period column was set to Date type.



Period	Accomodation type	Travel	Tourists
1/1/2012	Hotel	Domestic	735000
2/1/2012	Hotel	Domestic	749000
3/1/2012	Hotel	Domestic	889000
4/1/2012	Hotel	Domestic	969000
5/1/2012	Hotel	Domestic	1060000
6/1/2012	Hotel	Domestic	1032000
7/1/2012	Hotel	Domestic	1009000
8/1/2012	Hotel	Domestic	1106000
9/1/2012	Hotel	Domestic	1045000
10/1/2012	Hotel	Domestic	1025000
11/1/2012	Hotel	Domestic	859000
12/1/2012	Hotel	Domestic	810000
1/1/2013	Hotel	Domestic	698000
2/1/2013	Hotel	Domestic	744000
3/1/2013	Hotel	Domestic	881000
4/1/2013	Hotel	Domestic	958000
5/1/2013	Hotel	Domestic	1078000
6/1/2013	Hotel	Domestic	1088000
7/1/2013	Hotel	Domestic	1067000
8/1/2013	Hotel	Domestic	1139000

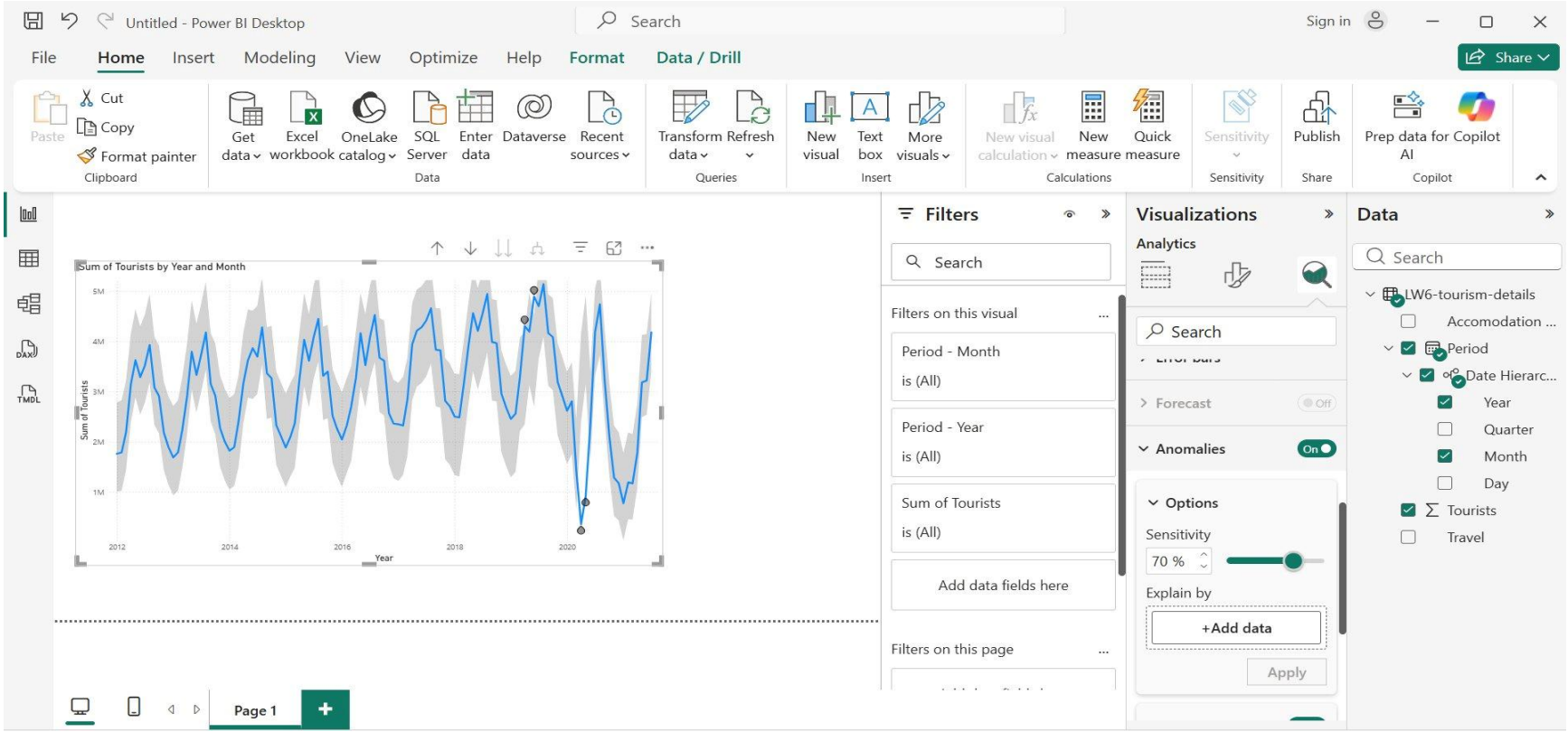
Part 2 — Line Chart: Sum of Tourists by Month

A line chart was created with Period on the X-axis and Tourists on the Y-axis (monthly view).



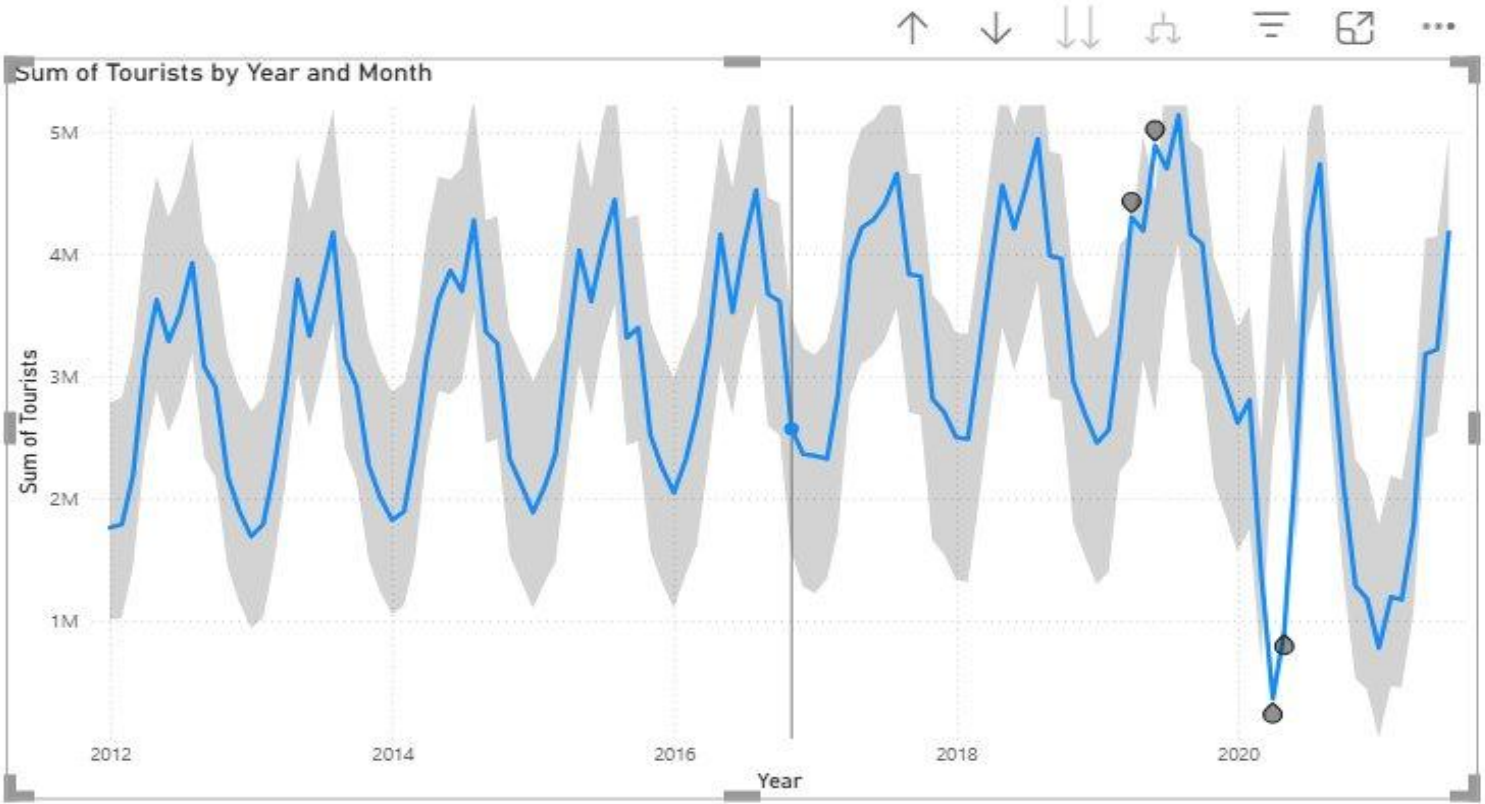
Part 3 — Anomaly Detection Enabled (70% Sensitivity)

Anomalies toggle turned ON in the Analytics pane. Sensitivity set to 70%. Shaded bands show expected range.



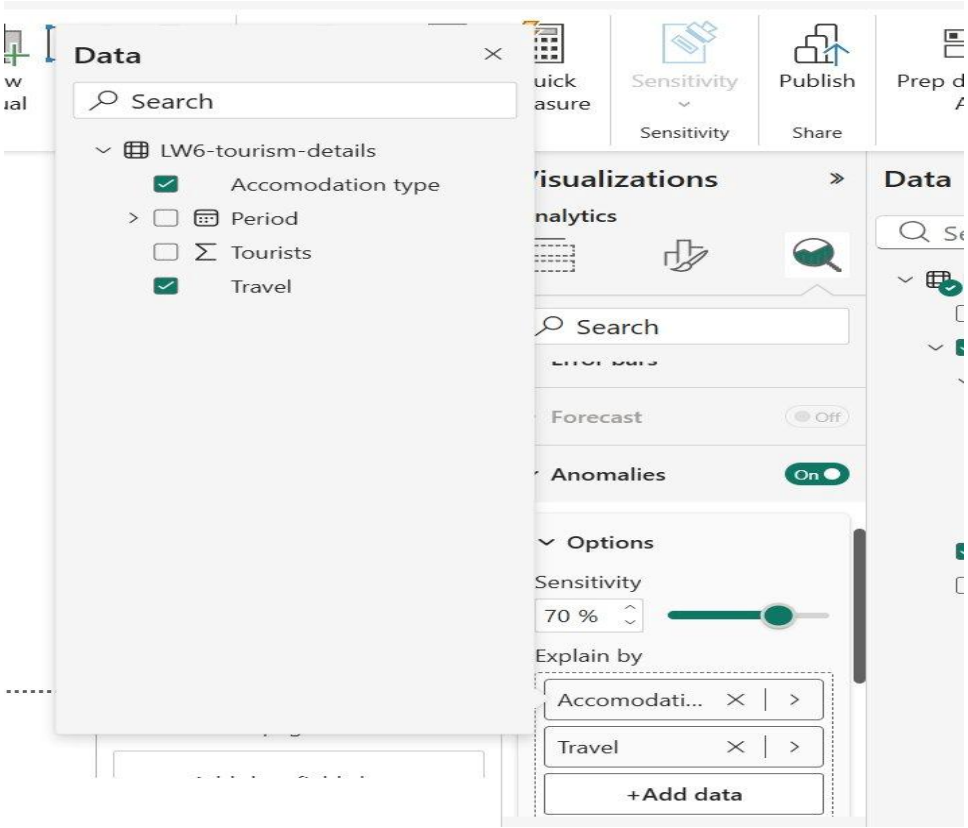
Anomaly Points Highlighted (2019–2021)

Anomaly markers appear around 2019 (unexpectedly high) and 2020–2021 (sharp drops — COVID-19 impact).



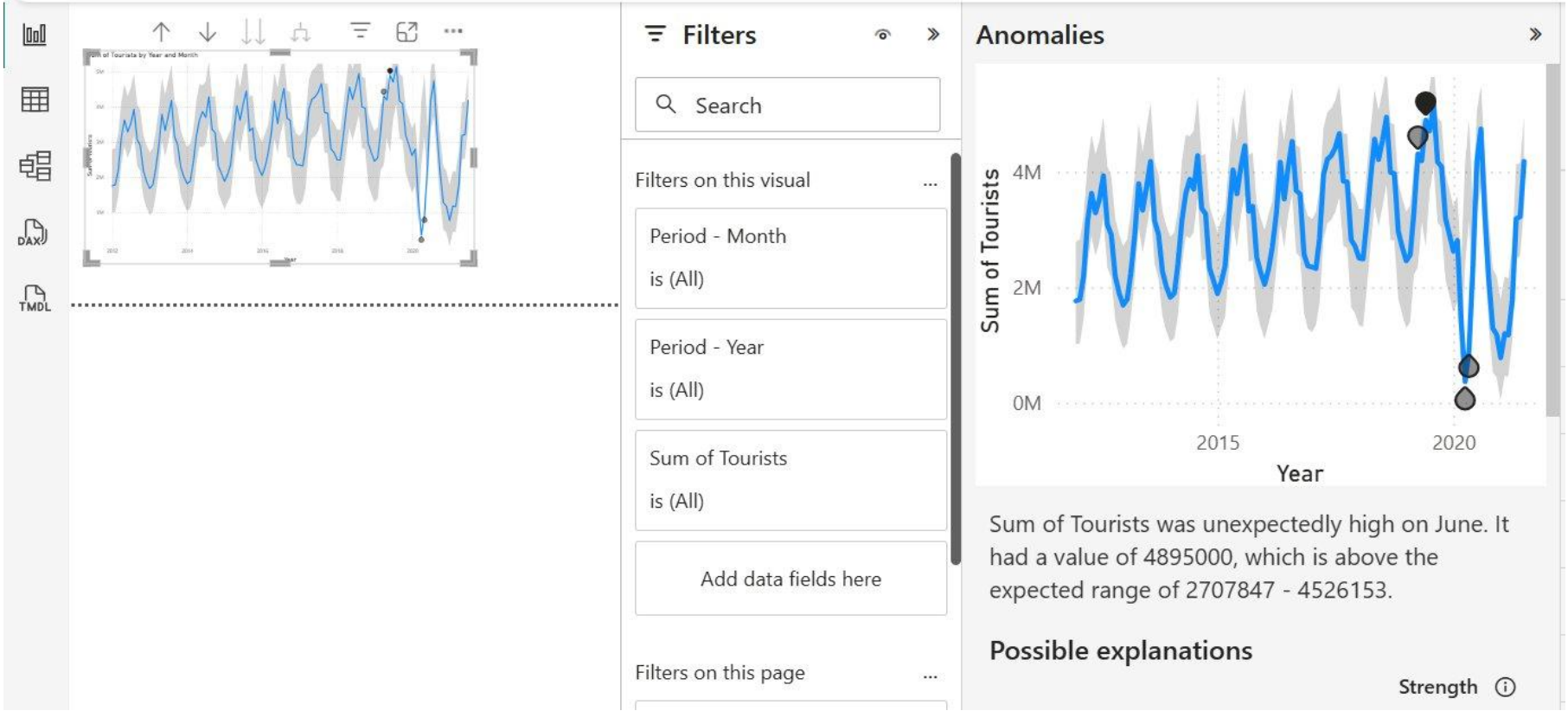
Part 4 — Explain By: Accommodation Type & Travel

'Accommodation type' and 'Travel' were dragged into the Explain By section to provide possible explanations.



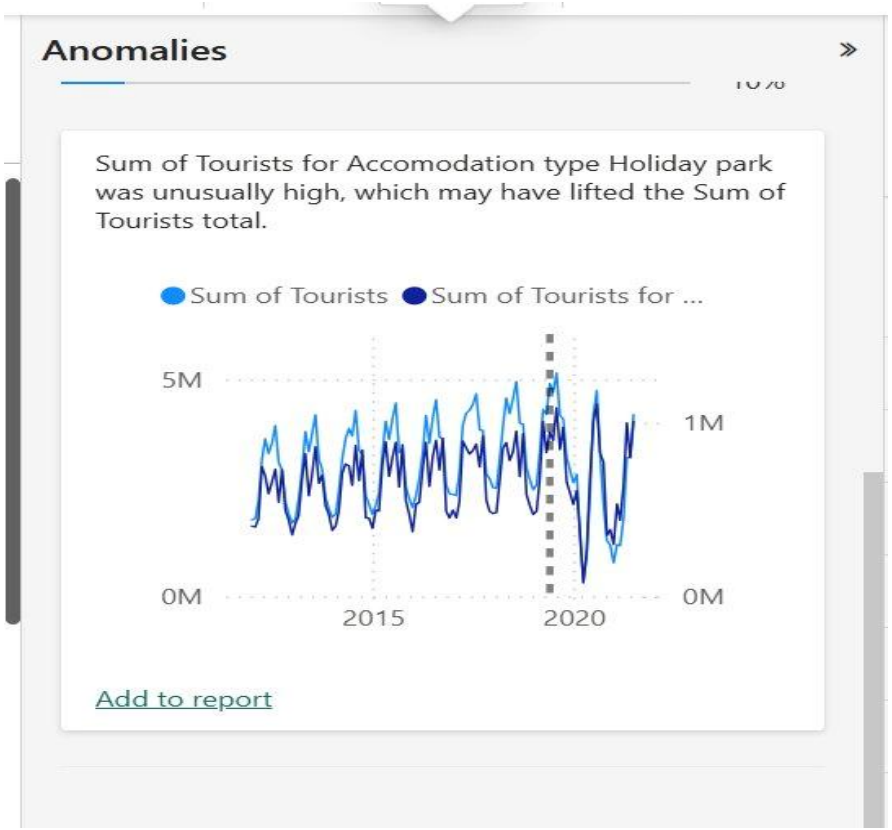
Anomaly: Unexpectedly HIGH — June (4,895,000 Tourists)

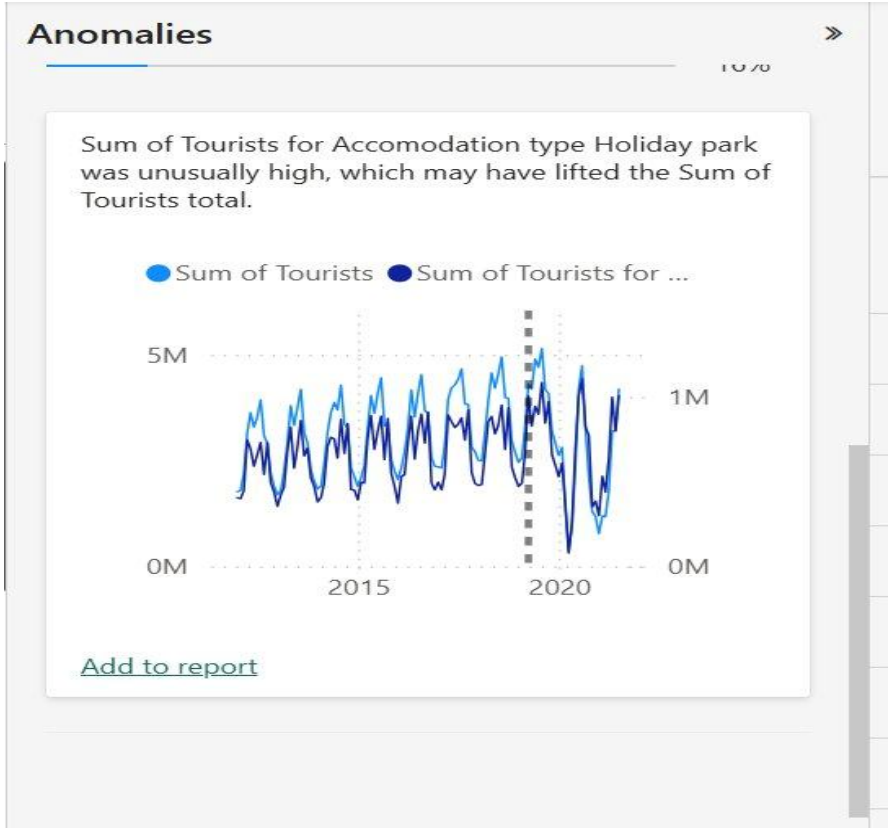
Value of 4,895,000 exceeded the expected range of 2,707,847 – 4,526,153. Triggered the anomaly flag.



Explanation: Holiday Park — Unusually High

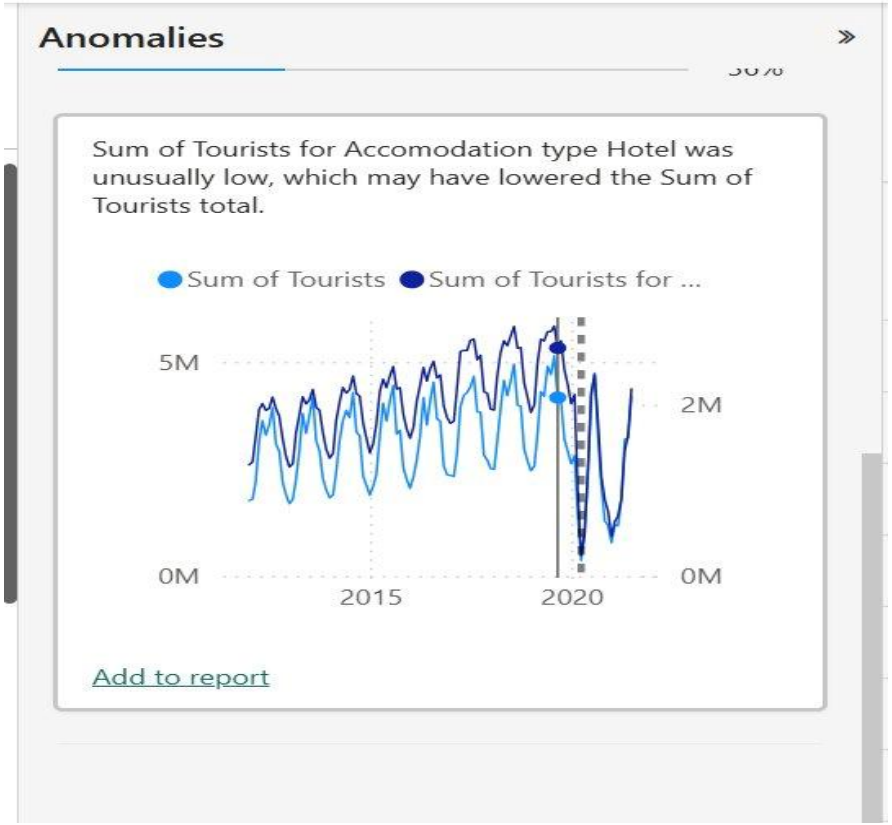
Power BI suggested that the 'Holiday Park' accommodation type was unusually high and may have lifted the total.

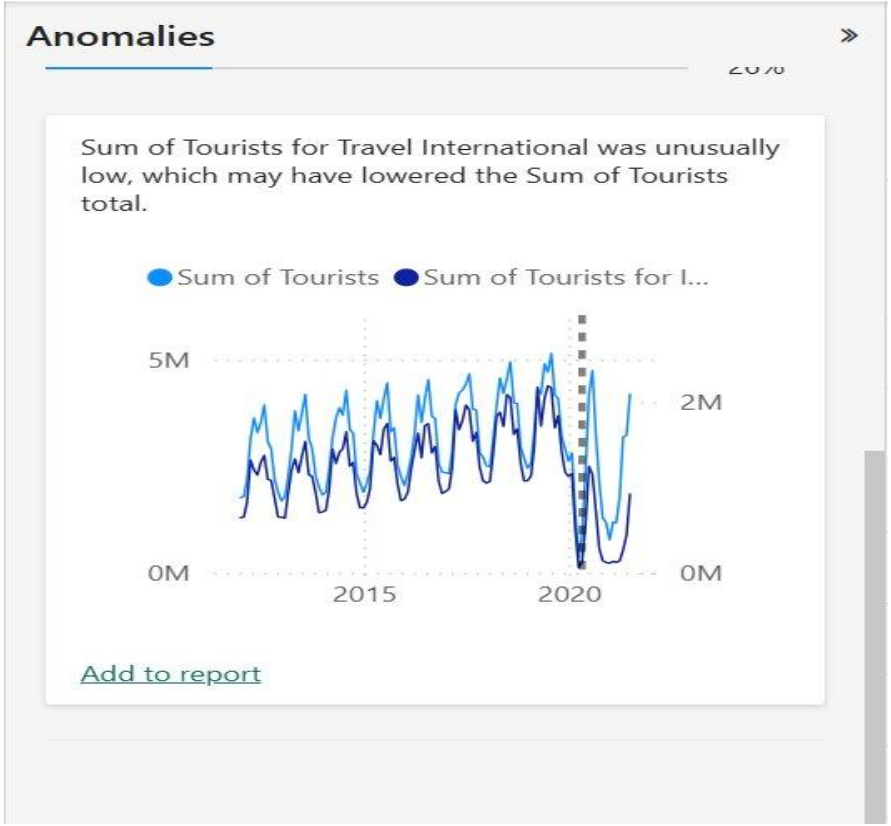




Explanations: Hotel (Low) & International Travel (Low)

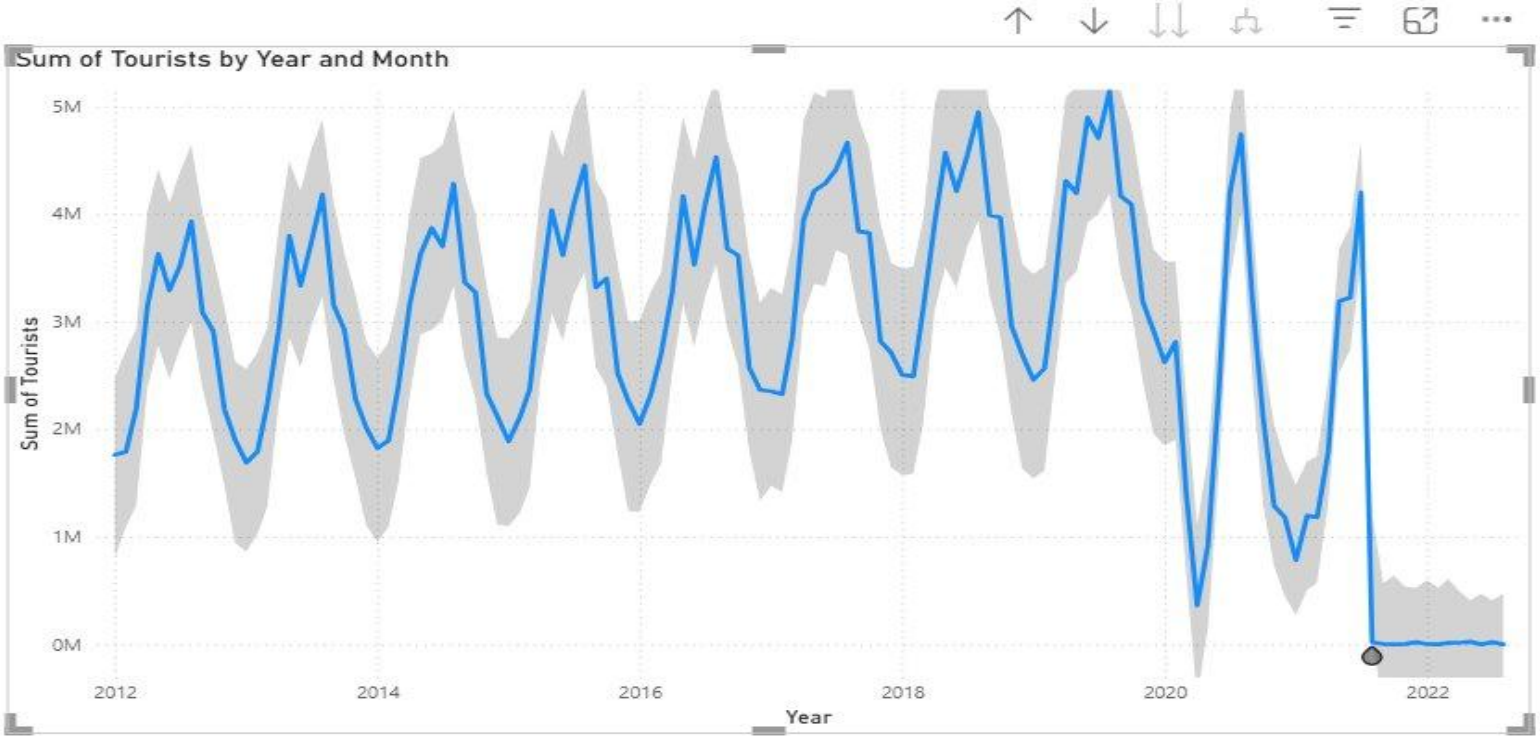
Both 'Hotel' accommodation and 'International' travel were identified as unusually low contributors.





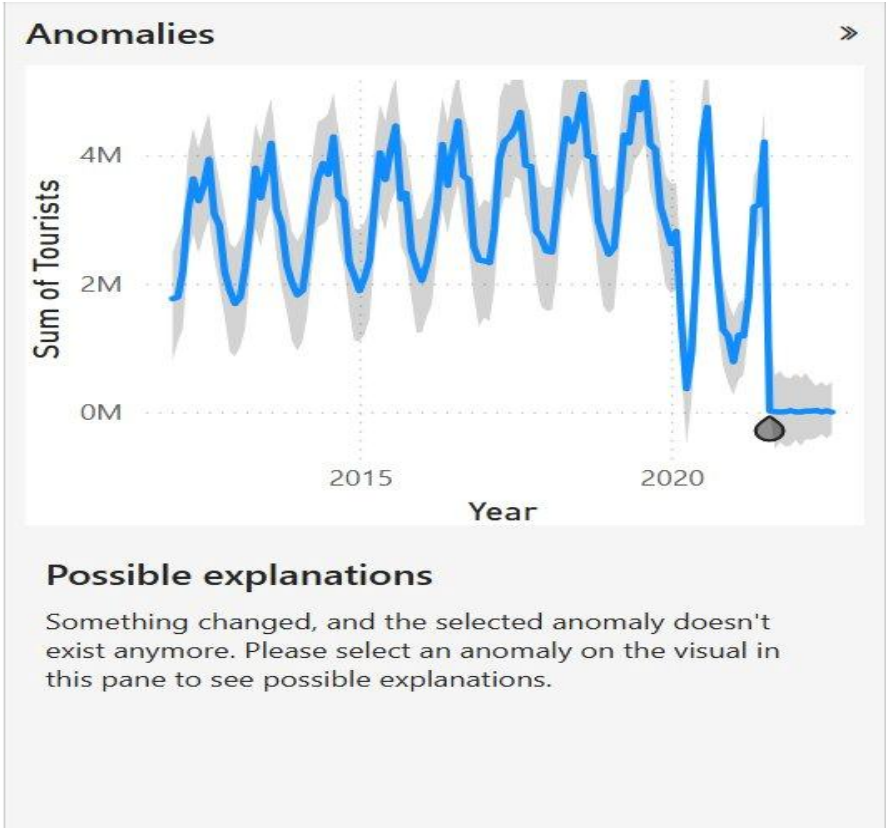
Part 6 — After Adding the New Anomaly Dataset

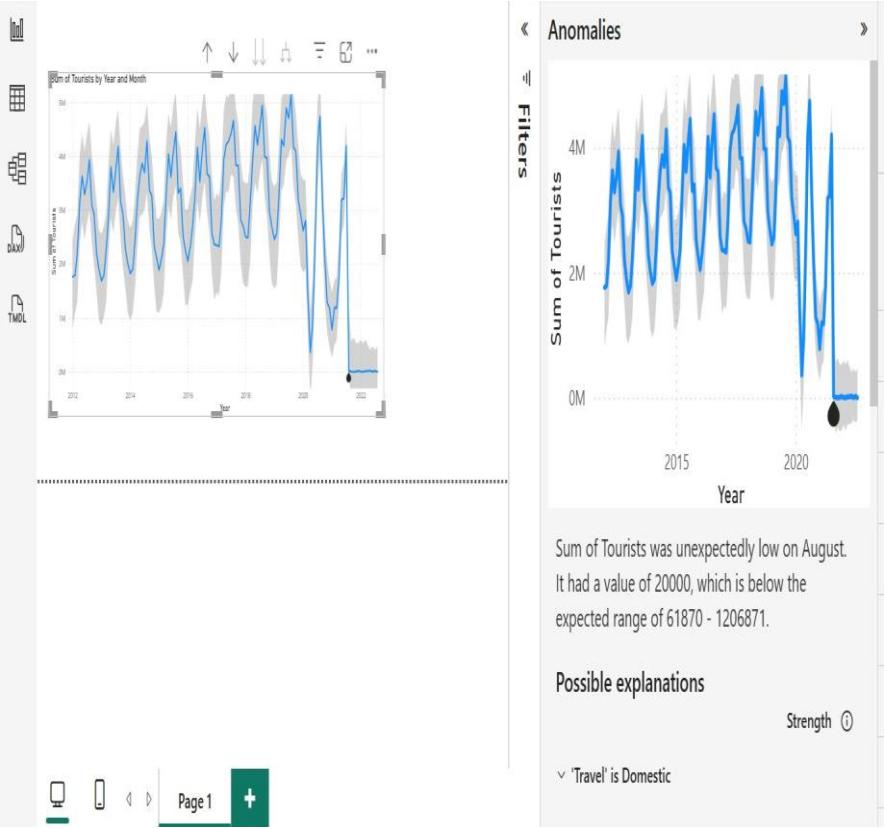
New 20-row dataset (2021–2022) appended. Report refreshed — new anomaly marker detected in early 2022.



New Anomaly — August 2022: Unexpectedly LOW (20,000)

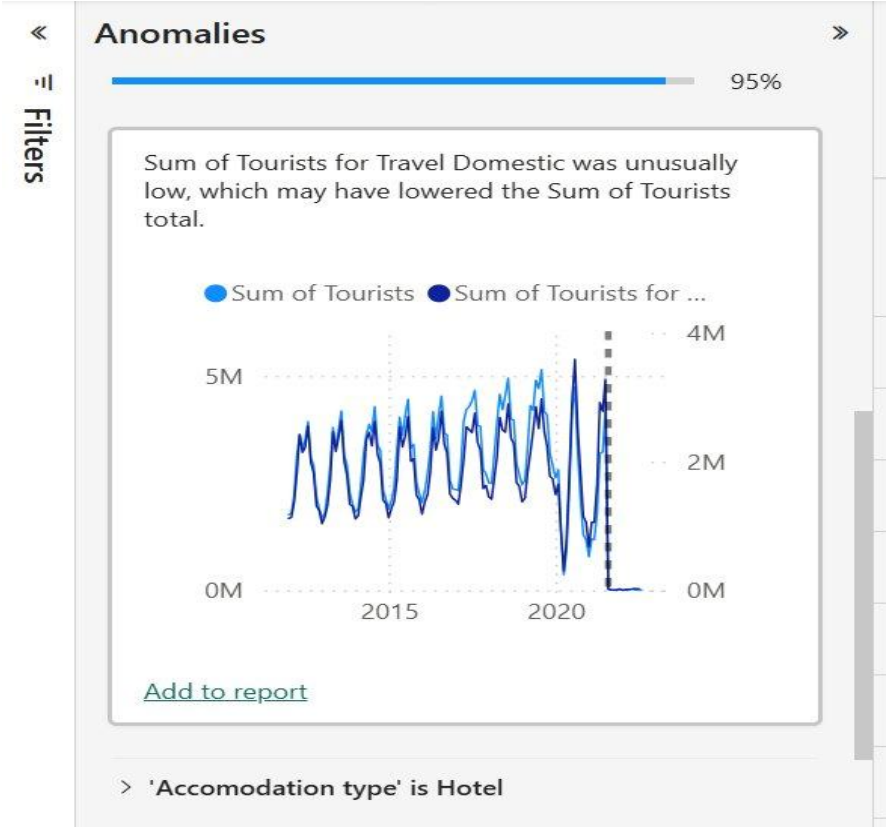
Value of 20,000 tourists fell far below the expected range of 61,870 – 1,206,871. Domestic travel was the main explanation.

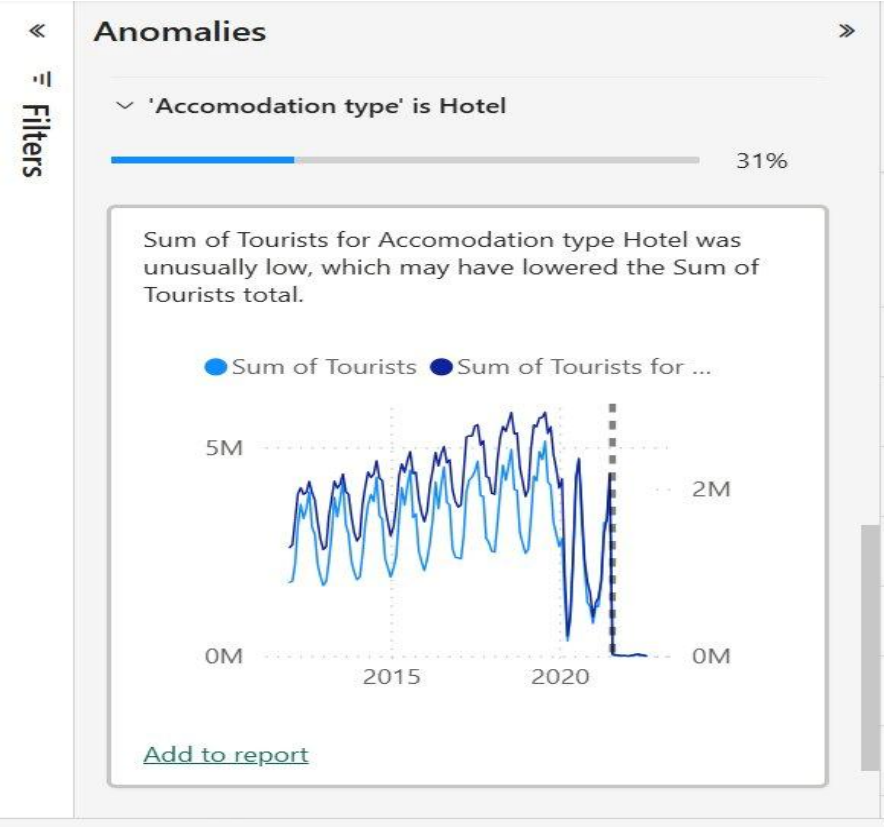




Explanations: Domestic Travel (95%) & Hotel (31%)

'Travel is Domestic' had the strongest explanation strength (95%), followed by 'Accommodation type is Hotel' (31%).





Part 5 — Guide Questions & Analysis

1. What trend and seasonality patterns are observed?

Tourist numbers show consistent seasonal peaks mid-year (June–August) and dips in winter months. An overall upward trend from 2012 to 2019 is visible.

2. Which anomalies are obvious vs. subtle?

The 2020 crash (COVID-19) is highly obvious — a dramatic drop far outside the expected band. The June 2019 peak is subtle — it only slightly exceeds the upper bound.

3. What factors contributed to anomalies?

The high anomaly in 2019 was linked to unusually high Holiday Park bookings. The 2020–2022 lows were driven by COVID-19 — especially drops in International travel and Hotel stays.

4. Are the AI explanations logically valid?

Yes. The suggestions align with real-world context. COVID-19 restrictions directly impacted hotel stays and international travel, confirming the AI's explanations are credible.

Summary

- The tourism dataset exhibited consistent seasonal trends with peaks in mid-year months.
- Anomaly detection at 70% sensitivity successfully flagged both high and low outliers.
- The June 2019 anomaly was driven by an unusually high number of Holiday Park visitors.
- The 2020–2022 anomalies clearly reflect the impact of COVID-19 — sharp drops in Hotel stays and International travel.
- After appending the new 20-row dataset, a new low anomaly in August 2022 was detected with 'Domestic' travel as the primary explanation (95% strength).
- The AI-generated explanations were logically valid and consistent with known global events.